

UNIT - I

1

Introduction to Cloud Computing

Syllabus

*Importance of Cloud Computing, Characteristics, Pros and Cons of Cloud Computing, Migrating into the Cloud, Seven-step model of migration into a Cloud, Trends in Computing. **Cloud Service Models** : SaaS, PaaS, IaaS, Storage. **Cloud Architecture** : Cloud Computing Logical Architecture, Developing Holistic Cloud Computing Reference Model, Cloud System Architecture, Cloud Deployment Models.*

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- 1.5 Trends in Computing
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1.1 Overview of Cloud Computing

SPPU : March-19

- Idea of cloud computing was introduced by computer scientist John McCarthy publicly in 1961.
- Then in 1969, Leonard Kleinrock, a chief scientist of the ARPANET project comments about Internet.
- The general public has been leveraging forms of Internet-based computer utilities since the mid-1990s through various incarnations of search engines, e-mail services, open publishing platforms and other types of social media.
- Though consumer-centric, these services popularized and validated core concepts that form the basis of modern-day cloud computing.
- The Salesforce.com provides remote service from 1999 to organization. Amazon launched its web services in 2002 and it provides services to organization for storage and remote computing.
- Cloud computing definition as per Gartner "a style of computing in which scalable and elastic IT-enabled capabilities are delivered as a service to external customers using Internet technologies".
- In 2008, Gartner's original definition of cloud was changed. In the definition, "massively scalable" was used instead of "scalable and elastic."
- **NIST definition of cloud :** Cloud computing is a pay-per-use model for enabling available, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort or service-provider interaction.
- The above cloud definition was published by NIST in 2009, followed by a revised version after further review and industry input that was published in September of 2011.
- Cloud computing refer to a variety of services available over the Internet that deliver compute functionality on the service provider's infrastructure.
 - Its environment (infrastructure) may actually be hosted on either a grid or utility computing environment, but that doesn't matter to a service user.
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Cloud Computing = Software as a service + Platform as a service
+ Infrastructure as a service + Data as a service

- Cloud computing is a general term used to describe a new class of network based computing that takes place over the Internet, basically a step on from utility computing.
- In other words, this is a collection/group of integrated and networked hardware, software and Internet infrastructure (called a platform).
- Fig. 1.1.1 shows cloud symbol. It denotes cloud boundary.
- Using the Internet for communication and transport provides hardware, software and networking services to clients.
- These platforms hide the complexity and details of the underlying infrastructure from users and applications by providing very simple graphical interface or API.
- In addition, the platform provides on demand services that are always on anywhere, anytime and anyplace. Pay for use and as needed, elastic.
- The hardware and software services are available to the general public, enterprises, corporations and business markets.

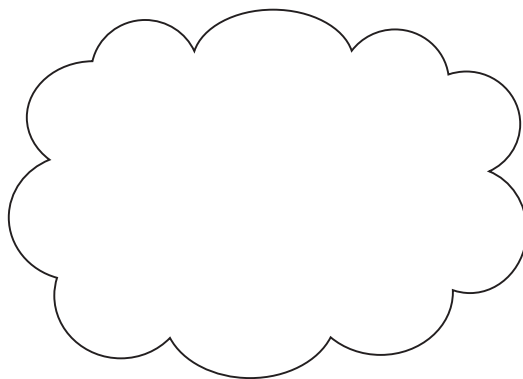


Fig. 1.1.1 Cloud symbol

1.1.1 Cloud Components

- Cloud computing solutions are made up of several elements. Fig. 1.1.2 shows cloud components.
 1. **Clients** : Mobile, terminals or regular computers.
 2. **Benefits** : Lower hardware costs, lower IT costs, security, data security, less power consumption, ease of repair or replacement, less noise.
 3. **Data centers** : Collection of servers where the application to subscribe is housed. It could be a large room in the basement of your building or a room full of servers on the other side of the world
 4. **Virtualizing servers** : Software can be installed allowing multiple instances of virtual servers to be used and a dozen virtual servers can run on one physical server.
 5. **Distributed servers** : Servers don't all have to be housed in the same location. It can be in geographically disparate locations. If something were to happen at one site, causing a failure, the service would still be accessed through another site. If the cloud needs more hardware, they can add them at another site.

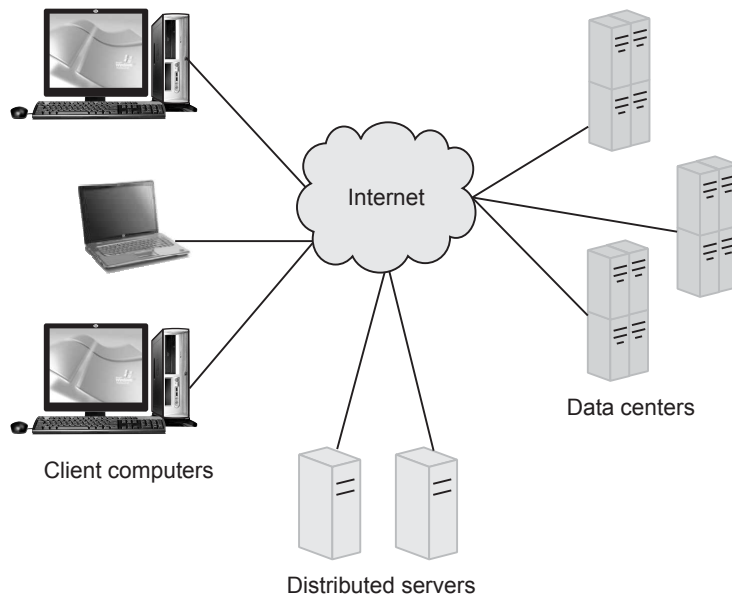


Fig. 1.1.2 Cloud components

Review Question

1. What is cloud computing ? What are the components of cloud computing ? Explain with neat diagram.

SPPU : March-19, In Sem, Marks 5

1.2 Importance of Cloud Computing

SPPU : March-19

- It is clear that utilizing the cloud is a trend that continues to grow. It provides better data storage, data security, flexibility, increased collaboration between employees, and changes the workflow of small businesses and large enterprises to help them make better decisions while decreasing costs.
- The significance of the cloud is increasing exponentially. Gartner forecasts that the cloud services market will grow 17.3 % in 2019 (\$206.2 billion) and by 2022, 90 % of organizations will be using cloud services.
- Considering all the potential and development cloud computing has undergone in recent years, there are also many challenges businesses are facing.
- Cloud storage works by storing data on remote servers, where it can be maintained, managed, backed up and accessed remotely. Data stored in the cloud is accessible by any device at any time, as long as permissions are in place. Despite its accessibility, data stored via the cloud is extremely safe and secure.

- Cloud computing offers the media streaming companies a cost effective, flexible, secure and scalable solution for setting up their work-flow. This serves to meet the media streaming companies requirements for both video on demand and live streaming services.
- During a live broadcast of India v/s Pakistan match, the traffic is easily 10-20 times that on normal days. Provisioning for servers and other infrastructure would end up creating a large overlay of upfront cost for infrastructure which otherwise would not be utilised in normal days. The cloud with its pay per use strategy offers economies of scale with effective costing strategies at a very high rate of reliability and security.

1.2.1 Characteristics of Cloud Computing

1. **On-demand self-service** : A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed without requiring human interaction with each service's provider.
2. **Ubiquitous network access** : Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms.
3. **Location-independent resource pooling** : The provider's computing resources are pooled to serve all consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
4. **Rapid elasticity** : Capabilities can be rapidly and elastically provisioned to quickly scale up, and rapidly released to quickly scale down.
5. **Pay per use** : Capabilities are charged using a metered, fee-for-service, or advertising-based billing model to promote optimization of resource use.

1.2.2 Pros and Cons of Cloud Computing

Pros of cloud computing :

1. **Lower computer costs** : Since applications run in the cloud, not on the desktop PC, your desktop PC does not need the processing power or hard disk space demanded by traditional desktop software.
2. **Improved performance** : Computers in a cloud computing system boot and run faster because they have fewer programs and processes loaded into memory.
3. **Reduced software costs** : Instead of purchasing expensive software applications, you can get most of what you need for free.

4. **Instant software updates** : When you access a web-based application, you get the latest version - without needing to pay for or download an upgrade.
5. **Improved document format compatibility** : You do not have to worry about the documents you create on your machine being compatible with other user's applications or operating systems.
6. **Unlimited storage capacity** : Cloud computing offers virtually limitless storage.
7. **Increased data reliability** : Unlike desktop computing, in which if a hard disk crashes and destroy all your valuable data, a computer crashing in the cloud should not affect the storage of your data.
8. **Universal document access** : All your documents are instantly available from wherever you are.
9. **Latest version availability** : The cloud always hosts the latest version of your documents; as long as you are connected, you are not in danger of having an outdated version.
10. **Easier group collaboration** : Sharing documents leads directly to better collaboration.
11. **Device independence** : Move to a portable device and your applications and documents are still available.

Cons of cloud computing :

1. It requires a constant Internet connection : Cloud computing is impossible if you cannot connect to the Internet.
2. Features might be limited.
3. Stored data might not be secure : With cloud computing, all your data is stored on the cloud.
4. Does not work well with low-speed connections.

1.2.3 Cloud Applications

1. Through cloud cost flexibility, online marketplace gains access to more powerful analytics online. Cloud takes away the need to fund the building of hardware, installing software or paying dedicated software license fees.
2. Greater business scalability enables online video retailer to meet spikes in demand : Cloud enables businesses not just IT operations to add or provision computing resources just at the time they're needed.
3. Greater market adaptability provides online entertainment platform the ability to reach any type of customer device. A third of the executives we surveyed believe

cloud can help them adapt to diverse user groups with a diverse assortment of devices.

4. Masked complexity enables access to services, no matter how intricate the technology they're built on.
5. With context-driven variability, "intelligent assistants" are possible. "Because of its expanded computing power and capacity, cloud can store information about user preferences, which can enable product or service customization," the report states.
6. Ecosystem connectivity enables information exchange across business partners.

Review Question

1. What are the advantages and disadvantages of cloud computing ?

SPPU : March-19, In Sem, Marks 5

1.3 Migrating into the Cloud

- Cloud migration is the process of transferring data, application code and other technology-related business processes from an on-premise or legacy infrastructure to the cloud environment.
- Cloud migration is a phenomenal transformation in the business information system domain as it provides adequate services for the growing needs of businesses. However, moving data to the cloud requires preparation and planning in deciding on an approach.
- Technology companies providing cloud and managed services should work closely with businesses and support them on three main aspects as part of their cloud migration journey :
 - a) Developing the right approach by ensuring clarity on the desired business outcome or end result.
 - b) Taking advantage of relevant talent and expertise to address companies' core business issues.
 - c) Using appropriate toolsets that are user-friendly, secure and viable in the long term.
- Your cloud computing migration strategy must deliver elasticity, agility, and scalability to help your organization take advantage of emerging opportunities and pivot to address business and industry changes.
- Achieving these goals will broadly employ a mix of Software as a Service (SaaS), Infrastructure as a Service (IaaS) and Platform as a Service (PaaS) cloud

architecture layers. You may employ these across hybrid and multi-cloud environments based on business needs and best practices.

- Cloud migration strategy should focus on four broad areas :
 - a) Security that determines application and data access controls, data security at rest and in transit and backup and data recovery.
 - b) Governance and compliance that makes sure you meet regulatory compliance and all aspects of data security to go beyond the limits of shared security models of cloud services providers.
 - c) Cost management that looks at immediate and long-term cost savings that guide where and how you migrate, manage, and monitor applications and workloads.
 - d) Accessibility, resilience, and scalability that are driven by customer and workforce UX needs, service enhancements and innovation.
- Benefits of migrating to the cloud include :
 - a) Increased agility and flexibility
 - b) Ability to innovate faster
 - c) Easing of increasing resource demands
 - d) Better managing of increased customer expectations
 - e) Reduction in costs
 - f) Deliver immediate business results
 - g) Simplify IT
 - h) Shift to everything as-a-service
 - i) Better consumption management
 - j) Cloud scalability
 - k) Improved performance

Cloud migration process :

- **Cloud migration process is divided into three phases : *plan, execute (run) and monitor***
- 1. **Plan** : Cloud migration requires solid planning to be successful. It define following parameters :
 - a. Identify business objective
 - b. Identify key business drivers
 - c. Get executive sponsorship
 - d. Providing full visibility into your on-premise environment, including all system dependencies.

- f. Evaluating performance, server, and security requirements.
- 2. **Execute or run** : Once your environment has been assessed and a plan has been mapped out, it's necessary to execute your migration. The main challenge here is carrying out your migration with minimal disruption to normal operation, at the lowest cost, and over the shortest period of time.
- 3. **Monitor** : After a successful migration, tools and processes should be implemented to monitor the new cloud environment.
- A formal issue tracking process should be created to ensure that everyone impacted by the migration has a way to report problems, and IT has a simple way to manage all requests.

Challenges of cloud migration :

1. **Interoperability** : It is no easy feat to get existing applications to communicate with newer cloud environments.
2. **Resource availability** : The migration process might require taking in-house servers temporarily offline. But downtime could be disastrous to application performance, thus customer loyalty, if not supported by a proper plan for disaster recovery.
3. **Data integrity** : How will user keep data secure while moving it to the cloud where have less control ?
4. **Resource management** : Not all IT professionals trust the cloud yet. If team was used to managing physical servers, they might need educating on the new infrastructure or even reconfiguring to introduce new roles.

1.4 Seven-step Model of Migration into a Cloud

- Seven-step model is as follows :
 1. Different variances of cloud services
 2. Cloud as a tool
 3. Cloud compatible
 4. Current cost
 5. To manage rather than operate
 6. To simplify
 7. Gain more knowledge.
- **Step 1** : Cloud migration assessments comprise assessments to understand the issues involved in the specific case of migration at the application level or the code, the design, the architecture, or usage levels.

These assessments are about the cost of migration as well as about the ROI that can be achieved in the case of production version.

- **Step 2** : An isolating all systemic and environmental dependencies of the enterprise application components within the captive data center.
- **Step 3** : Generating the mapping constructs between what shall possibly remain in the local captive data center and what goes onto the cloud.
- **Step 4** : Substantial part of the enterprise application needs to be rearchitected, redesigned, and reimplemented on the cloud.
- **Step 5** : We leverage the intrinsic features of the cloud computing service to augment our enterprise application in its own small ways.
- **Step 6** : We validate and test the new form of the enterprise application with an extensive test suite that comprises testing the components of the enterprise application on the cloud as well.
- **Step 7** : Test results could be positive or mixed. In the latter case, we iterate and optimize as appropriate. After several such optimizing iterations, the migration is deemed successful.

1.5 Trends in Computing

1. IT maturity analysis

- The technology maturity analysis, which compares the technologies against each other in terms of their adoption rates and current investment rates.
- Major trends that emerged in cloud computing technology are small and medium business, hybrid cloud model etc.

2. Technology Trends to Watch

- Virtualization is technology that lets user create useful IT services using resources that are traditionally bound to hardware.
- Virtualization is composed of server, storage, desktop, application, infrastructure and network. It is compatible with today's technology.
- Data growth : Day by day, data may goes on increasing and organization can not provide storage. So cloud is solution for storing and processing of growing data.
- Green IT : The term green IT means nothing other than the use of information technology in a manner that preserves resources and protects the environment. Cloud computing, the provision of IT infrastructure via the Internet, plays an important role in green IT. Cloud allows the replacement of poorly utilized data centers with highly optimized server environments. Besides reducing the energy costs, it also minimizes the time required to maintain a company's proprietary systems.

1.6 Cloud Service Models

SPPU : March-19

- Service models describe the type of service that the service provider is offering. The best-known service models are Software as a Service, Platform as a Service, and Infrastructure as a Service.
- The service models build on one another and define what a vendor must manage and what the client's responsibility is.
- Service models : This consists of the particular types of services that you can access on a cloud computing platform.
- Cloud service is any service made available to users on demand via the Internet from a cloud computing provider's servers as opposed to being provided from a company's own on-premises servers.
- Cloud services are designed to provide easy, scalable access to applications, resources and services, and are fully managed by a cloud services provider.
- A cloud service can exist as a simple web-based software program with a technical interface invoked via the use of a messaging protocol, or as a remote access point for administrative tools or larger environments and other IT resources.
- The organization that provides cloud-based IT resources is the cloud provider. Cloud providers normally own the IT resources for lease by cloud consumers, and could also resell IT resources leased from other providers.
- Cloud computing, often described as a stack, has a broad range of services built on top of one another under the name cloud.
- Fig. 1.6.1 shows cloud computing stack.
- Flavors of Cloud computing is as follows;
 1. SaaS applications are designed for end-users, delivered over the web
 2. PaaS is the set of tools and services designed to make coding and deploying those applications quick and efficient

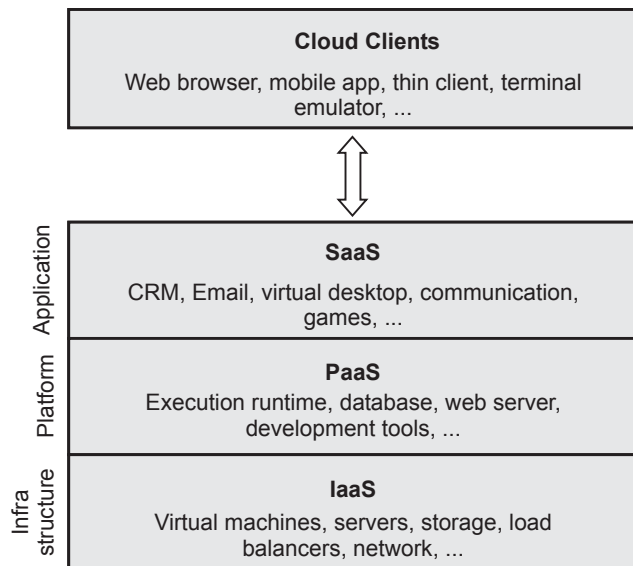


Fig. 1.6.1 Cloud computing stack

3. IaaS is the hardware and software that powers it all - servers, storage, networks, operating systems.

1.6.1 Software as a Service (SaaS)

- Model in which an application is hosted as a service to customers who access it via the Internet.
- The provider does all the patching and upgrades as well as keeping the infrastructure running.
- The traditional model of software distribution, in which software is purchased for and installed on personal computers, is referred to as product.
- In this model, the user, client or consumer runs an application from a cloud infrastructure. Through an interface such as a web browser, the client or user may access this application from a variety of devices.
- The complete application is offered as on demand service. This saves the client from having to invest in any software licenses or servers up front, and can save the provider money since they are maintaining and providing only a single application.
- In this model, the client does not manage cloud infrastructure, networks or servers, storage, or operating systems. Even, Microsoft, Google, and Zoho offer SaaS.
- The SaaS concept can be defined as providing robust "web-based, on-demand software, storage and various applications" to organizations.
- The SaaS model has emerged as an alternative to traditional one-time licensing for providing and maintaining the software needed by knowledge workers within organizations.
- Fig. 1.6.2 shows SaaS.

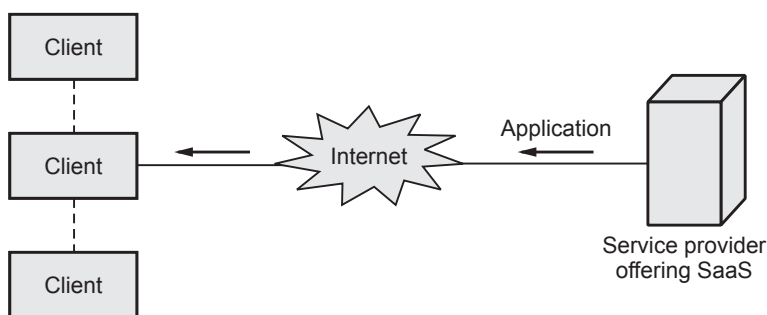


Fig 1.6.2 SaaS

Characteristics of SaaS :

1. Software applications or services are stored remotely.
2. A user can then access these services or software applications via the Internet.
3. In most cases, a user does not have to install anything onto their host machine, all they require is a web browser to access these services and in some cases, a browser may require additional plug-in/add-on for certain services.
4. Network-based management and access to commercially available software from central locations rather than at each customer's site, enabling customers to access applications remotely via the Internet.
5. Application delivery from a one-to-many model, as opposed to a traditional one-to-one model.

Benefits of SaaS :

1. You only pay for what you use
2. Easier administration and invoicing
3. Automatic updates and patch management
4. Compatibility : All users have access to the same version of software
5. Easier collaboration
6. It support automated update and patch management services.

1.6.2 Platform as a Service (PaaS)

- Platform as a service is another application delivery model and also known as **cloud-ware**. Supplies all the resources required to build applications and services completely from the Internet, without having to download or install software.
- Services include : Application design, development, testing, deployment, and hosting, team collaboration, web service integration, database integration, security, scalability, storage, state management, and versioning.
- PaaS is closely related to SaaS but delivers a platform from which to work rather than an application to work with.
- This model involves software encapsulated and offered as a service, from which higher levels of service may then be built. The user, customer, or client in this model is the one building applications which then run on the provider's infrastructure.
- This in turn provides customers and clients with the capability to deploy applications onto the cloud infrastructure using programming tools and languages, which the provider supports.

- The customer still does not manage the framework, network, servers or operating system, but has control over deployed applications and sometimes over the hosting environment itself.
- Some examples of Platform as a Service include Google's App Engine or Force.com
- PaaS consists of following components :
 1. Browser based development studio
 2. Pay contrary to billing
 3. Management and supervising tools
 4. Seamless deployment to host run time environment.
- **Characteristics of PaaS :**
 1. It support multi-tenant architecture.
 2. It support for development of group collaboration.
 3. PaaS systems can be deployed as public cloud services or as private cloud services.
 4. Provision of runtime environments. Typically each runtime environment supports either one or a small set of programming languages and frameworks
 5. Support for custom applications. Support for the development, deployment and operation of custom applications.
 6. Preconfigured capabilities. Many PaaS systems are characterized by capabilities that are preconfigured by the provider, with a minimum of configuration available to developers and customer operations staff.
 7. Support for porting existing applications. While many PaaS systems are primarily designed to support "born on the cloud" applications.
 8. Security is an important characteristic in PaaS. It needs to provide authentication and authorization to differentiate the access rights of different users

Benefits of Paas :

1. Scalability including rapid allocation and deallocation of resources with a pay-as-you-use model
2. Reduced capital expenditure
3. Reduced lead times with on-demand availability of resources
4. Self-service with reduced administration costs
5. Reduced skill requirements
6. Support of team collaboration
7. Ability to add new users quickly.

1.6.3 Infrastructure as a Service (IaaS)

- IaaS gives the storage room likeness to the in-house datacenter stood out from various organizations sorts.
- Center datacenter framework segments are capacity, servers (registering units), the system itself, and administration apparatuses for foundation upkeep and checking.
- Each of these parts has made a different market specialty. While some little organizations have practical experience in just a single of these IaaS cloud specialties, vast cloud suppliers like Amazon or Right Scale have offerings over all IaaS territories.
- Fig. 1.6.3 shows IaaS.

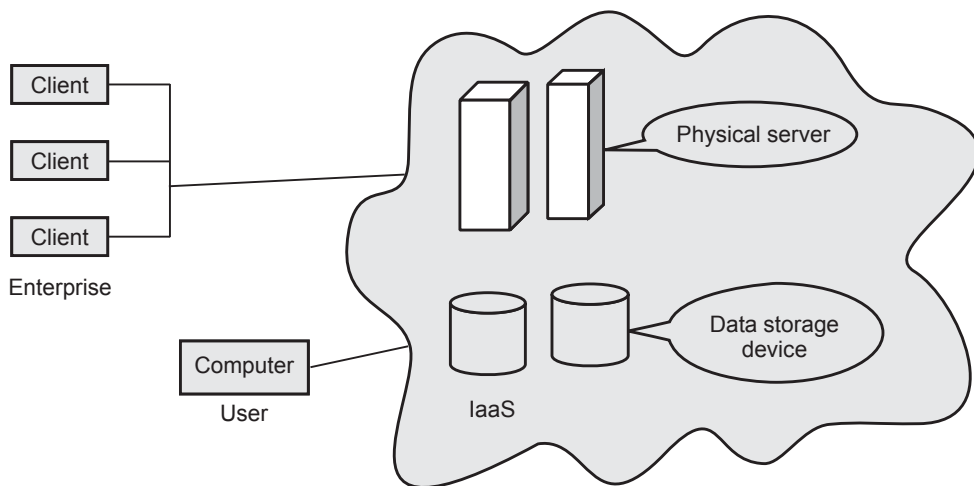


Fig. 1.6.3 IaaS

- It offers the hardware so that your organization can put whatever they want onto it. Rather than purchase servers, software, racks, and having to pay for the datacenter space for them, the service provider rents those resources :
 1. Server space
 2. Network equipment
 3. Memory
 4. CPU cycles
 5. Storage space
- Again, the customer is not managing cloud infrastructure, but in this case, the customer does control operating systems, deployed applications, storage, and sometimes-certain networking components
- Examples : Amazon EC2, Rackspace Mosso, GoGrid
- IaaS server types :
 1. **Physical server** : Actual hardware is allocated for the customer's dedicated use.
 2. **Dedicated virtual server** : The customer is allocated a virtual server, which runs on a physical server that may or may not have other virtual servers.

- 3. Shared virtual server :** The customer can access a virtual server on a device that may be shared with other customers.

Advantages of IaaS :

1. Elimination of an expensive and staff-intensive data center
2. Ease of hardware scalability
3. Reduced hardware cost
4. On-demand, pay as you go scalability
5. Reduction of IT staff
6. Suitability for ad hoc test environments
7. Allows complete system administration and management
8. Support multiple tenants

1.6.4 Difference between IaaS, PaaS and SaaS

IaaS	PaaS	SaaS
IaaS gives users automated and scalable environments	PaaS provides a framework for quickly developing and deploying applications	SaaS makes applications available through the internet.
Amazon Web Services, for example, offers IaaS through the Elastic Compute Cloud, or EC2	Google Cloud Platform provides another PaaS option in App Engine	SaaS applications such as Gmail, Dropbox, Salesforce, or Netflix
In IaaS, infrastructure as a service.	In PaaS, platform as a service	In SaaS, software as a service
Virtual platform on which required operating environment and application deployed	Operating environment was included.	Operating environment largely irrelevant, fully functional application provided
IaaS is a cloud service that provides basic computing infrastructure: servers, storage, and networking resources. In other words, IaaS is a virtual data center	PaaS refers to cloud platforms that provide runtime environments for developing, testing, and managing applications	SaaS allows people to use cloud-based web applications.
Major IaaS providers include Amazon Web Services, Microsoft Azure, and Google Compute Engine.	Examples of PaaS services are Heroku and Google App Engine.	email services such as Gmail and Hotmail are examples of cloud-based SaaS services.

IaaS services are available on a pay-for-what-you-use model	PaaS solutions are available with a pay-as-you-go pricing model.	SaaS services are usually available with a pay-as-you-go pricing model
Used by IT administrator	Used by software developers	Used by end user

Review Question

1. Explain benefits of SaaS in detail.

SPPU : March-19, In Sem, Marks 4

1.7 Cloud Computing Logical Architecture

- Fig. 1.7.1 shows architectural framework of cloud computing.

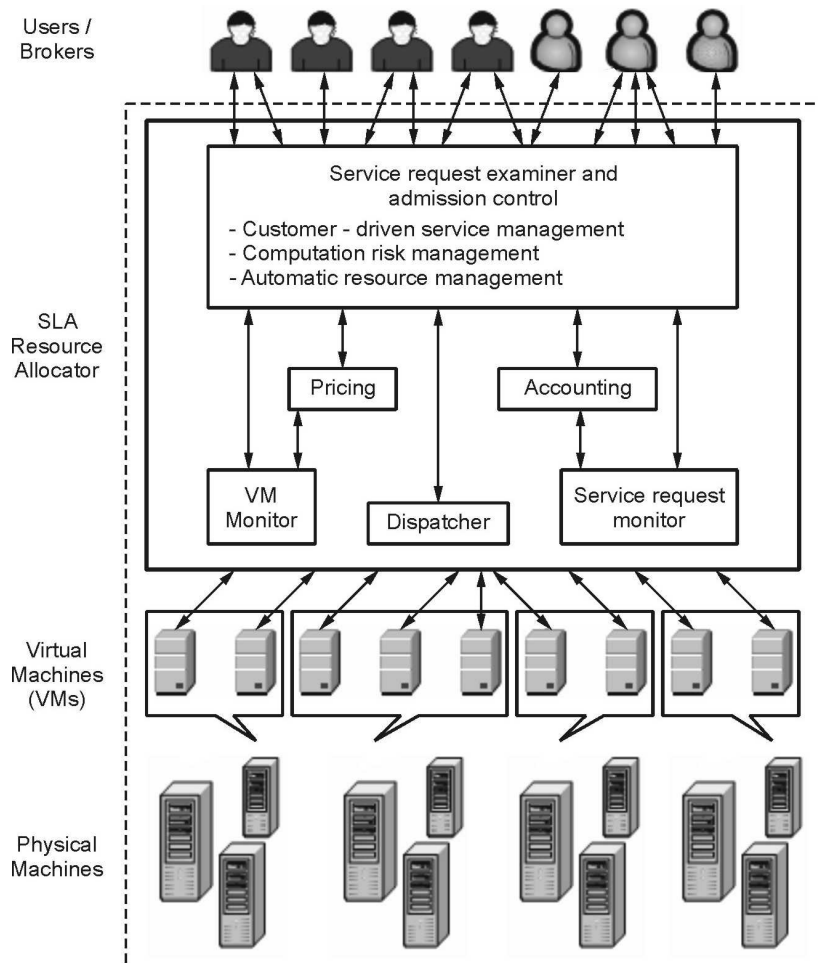


Fig. 1.7.1 Architectural framework

1. **Users/Brokers** : They submit their service requests from anywhere in the world to the cloud.
2. **SLA resource allocator** : It is a kind of interface between users and cloud service provider which enable the SLA-oriented resource management.
3. **Service request examiner and admission control** : It interprets the submitted request for QoS requirements before determining whether to accept or reject the request. Based on resource availability in the cloud and other parameters decide.
4. **Pricing** : It is in charge of billing based on the resource utilization and some factors. Some factors are request time, type etc.
5. **Accounting** : Maintains the actual usage of resources by request so that the final cost can be charged to the users.
6. **VM monitor** : Keeps tracks on the availability of VMs and their resources.
7. **Dispatcher** : The dispatcher mechanism starts the execution of admitted requests on allocated VMs.
8. **Service request monitor** : The request monitor mechanism keeps track on execution of request in order to be in tune with SLA.

Cloud computing service layers :

Parameters	Services	Description
Application Focused	Services	Services - Complete business services such as PayPal, OpenID, OAuth, Google Maps, Alexa
	Application	Application - Cloud based software that eliminates the need for local installation such as Google Apps, Microsoft Online
	Development	Development - Software development platforms used to build custom cloud based applications (PAAS and SAAS) such as Salesforce
Infrastructure Focused	Platform	Platform - Cloud based platforms, typically provided using virtualization, such as Amazon ECC, Sun Grid
	Storing	Storage - Data storage or cloud based NAS such as CTERA, iDisk, CloudNAS
	Hosting	Hosting - Physical data centers such as those run by IBM, HP, NaviSite, etc.

Cloud components :

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2. **Benefits** : Lower hardware costs, lower IT costs, security, data security, less power consumption, ease of repair or replacement, less noise.
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1.7.1 Cloud Computing Service Architecture

- Cloud computing services are divided into three classes :
 - 1) Infrastructure as a service,
 - 2) Platform as a service
 - 3) Software as a service.
- Fig. 1.7.2 shows the layered organization of the cloud stack from physical infrastructure to applications.

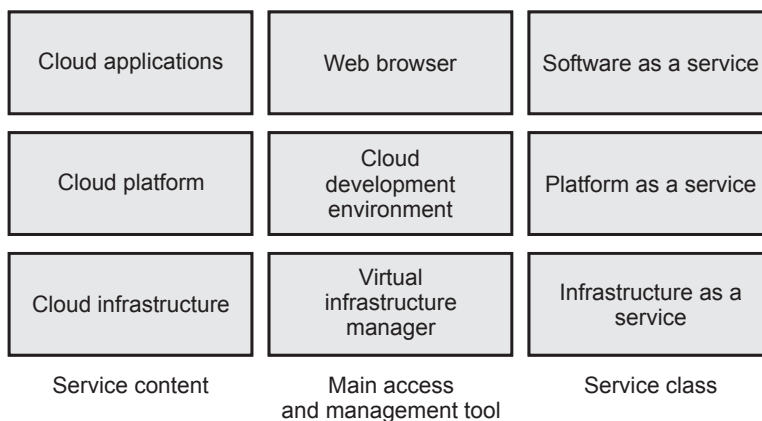


Fig. 1.7.2 Cloud stack

- Cloud services are designed to provide easy, scalable access to applications, resources and services and are fully managed by a cloud services provider.
- A cloud service can exist as a simple web-based software program with a technical interface invoked via the use of a messaging protocol or as a remote access point for administrative tools or larger environments and other IT resources.

- The organization that provides cloud-based IT resources is the cloud provider. Cloud providers normally own the IT resources for lease by cloud consumers, and could also resell IT resources leased from other providers.
- Cloud computing, often described as a stack, has a broad range of services built on top of one another under the name cloud.

1. Software as a service

- SaaS utilizes the internet to deliver applications, which are managed by a third-party vendor, to its users. A majority of SaaS applications run directly through your web browser, which means they do not require any downloads or installations on the client side.
- Salesforce.com uses SaaS model. SaaS applications are designed for end-users, delivered over the web.

2. Infrastructure as a service

- IaaS is a cloud computing service where enterprises rent or lease servers for compute and storage in the cloud. Users can run any operating system or applications on the rented servers without the maintenance and operating costs of those servers.
- Amazon Web Services mainly offers IaaS.
- IaaS is the hardware and software that powers it all - servers, storage, networks, operating systems.

3. Platform as a service

- Platform as a Service (PaaS), provide cloud components to certain software while being used mainly for applications. PaaS delivers a framework for developers that they can build upon and use to create customized applications.
- All servers, storage, and networking can be managed by the enterprise or a third-party provider while the developers can maintain management of the applications.
- Google AppEngine, an example of Platform as a Service.
- PaaS is the set of tools and services designed to make coding and deploying those applications quick and efficient.

1.8 Developing Holistic Cloud Computing Reference Model

- Cloud computing is an internet-based computing where software, shared resources and information are served to devices such as computers, electricity grid. In these computing techniques, how the work is getting done is hidden from its users. It describes a new consumption, delivery and supplement model for IT services.

- Cloud computing is one of the next significant stage in the Internet's evolution, providing the means through which everything from computing power to computing infrastructure, applications, business processes to personal collaboration can be delivered to the user as a service wherever and whenever they need.
- The "cloud" in cloud computing can be defined as the set of hardware, networks, storage, services and interfaces that combine to deliver aspects of computing as a service.
- Consumer cloud computing services has been well established ever since mainstream Internet. Known examples are Webmail services and social networking platforms.
- However the adoption of cloud computing within the enterprise sector has been slow. This slow uptake in cloud services that promises so much has been primarily influenced by the numerous security risks, concerns and challenges posed within such an environment.
- Governance, risk and compliance factors of cloud services need to be fully assessed and evaluated by organizations to provide informed judgments. Data and information lifecycle, source and origination, transfer, destination, validation and deletion all need to be understood.

1.8.1 Complete View of Cloud Management

- Despite significant attention and substantial efforts both in industry and academia, cloud computing has not yet reached its full potential. Commonly stated obstacles for cloud adoption include confusion about terminology with multiple delivery models (SaaS, PaaS, IaaS) and deployment cloud scenarios.
- The analysis of cloud computing obstacles in five high-level research challenges :
 1. Service life cycle optimization for improved construction, deployment and operation of cloud services.
 2. Adaptive self-preservation with increased autonomy in resource management, enabling fewer administrators to handle increasingly larger systems.
 3. Cloud and service self-management based on non-functional management criteria, extending beyond typical cost-performance trade-offs to also incorporate aspects of trust, risk and eco-efficiency in decision making.
 4. Support for service deployment and operation in multiple cloud architectures such as private clouds, cloud federation, cross-cloud service provisioning.
 5. Market and legislative issues including identification of new market roles and business models for clouds.

1.8.2 Cloud Computing Reference Model

- Cloud Computing Reference Model (CC-RM) is developed to facilitate the process of cloud modeling, deployment planning and architecture.
- The cloud computing reference model is an abstract model that characterizes and standardizes the functions of a cloud computing environment by partitioning it into abstraction layers and cross-layer functions.
- The cloud computing reference model will establish a cloud modeling and architecture foundation from which an organization can realistically plan, model, architect and deploy cloud computing in a pragmatic fashion to address real and pressing business and technical challenges.
- Cloud should not be treated as a solution looking for a problem but as a collection of cloud patterns that can be configured to meet a wide array of business and technical requirements.
- Elements of cloud computing reference model is as follows :
 - a. Cloud enablement model
 - b. Cloud deployment model
 - c. Cloud governance and operations model
 - d. Cloud ecosystem model.
- The cloud enablement model describes the tiers of cloud computing foundation, enablement and business capabilities provided by cloud platform and service providers to potential consumers of cloud-enabled technology and business capabilities.
- The Cloud Deployment Model (CDM) describes the range of cloud deployment scenarios available to the enterprise internal-private cloud, external public cloud, hybrid cloud and community clouds.
- Cloud governance and operations model : It describes the governance, security and privacy, operations and support, management and monitoring requirements for cloud computing and ensures that the user has considered all the potential operational risks for adopting cloud for their enterprises.
- The cloud ecosystem model considers the requirements of developing and sustaining a cloud ecosystem comprised of cloud providers, cloud consumers, cloud intermediaries, as well as the cloud network.
- Fig. 1.8.1 shows cloud computing reference model.

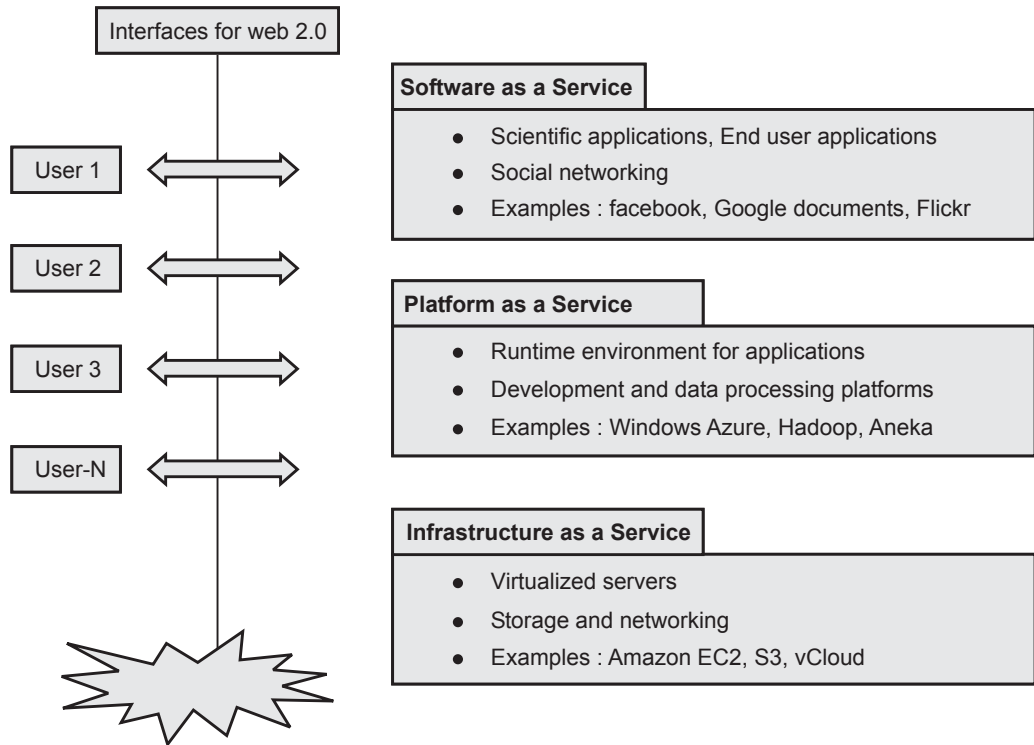


Fig. 1.8.1 Cloud computing reference model

Software as a Service

- SaaS provider dispose the applied software unified on their server, the user can subscribe applied software service from the manufacturer through Internet.
- Provider supply software pattern through Browser, and charge according to the quantity of software and using time. The advantage of this kind of service pattern is that the provider maintains and manages software, supplies the hardware facilities, the users can use software everywhere when they own the terminal which can log in Internet.

Platform as a Service (PaaS) :

- PaaS takes develop environment as a service to supply. This layer provides a platform for creating applications.
- PaaS solutions are essentially development platforms for which the development tool itself is hosted in the cloud and accessed through a browser.
- With PaaS, developers can build web applications without installing any tools on their computers and then deploy those applications without any specialized systems administration skills.

- Google App Engine is the representative product through their server and Internet.

Infrastructure as a Service (IaaS) :

- In this layer, servers, network devices, and storage disks are made available to organizations as services on a need-to basis.
- IaaS takes infrastructure which is made of many servers as a measurement service to the customers.
- It integrates memory and I/O devices, storage and computing ability into a virtual resources pool, and provides storage resources and virtualization service for the whole industry.
- This is a way of hosted hardware, and the customer pays when they use the hardware.
- For example, Amazon Web Service and IBM Blue Cloud all rent the infrastructure as a service.

1.9 Cloud System Architecture

- There are several factors that user need to take into consideration before designing own cloud-based systems architecture, particularly if user is considering a multi-cloud/region architecture.
 1. Speed - The cloud gives more flexibility to control the speed or latency of site/application.
 2. Complexity - Before constructing highly customized hybrid cloud solution architecture, make sure user properly understand the actual requirements of application, SLA, etc
 3. Cloud portability - Although it might be easier to use one of the cloud provider's tools or services, such as a load balancing or database service.
 4. Cost - Before launching servers, user should clearly understand the SLA and pricing models associated with cloud infrastructure(s)
 5. Security - Check security measurement provided by vendor.

Single Server

- A single server means only one server, whether physical or virtual, containing one web server, database, and application.
- LAMP stack is an example of single server. This structure is not widely utilized due to associated security risks.

1.9.1 Single Cloud Site Architectures

- Fig. 1.9.1 shows single cloud site architecture.
- Cloud contains load balancer server, application server and database.
- A single cloud environment is accomplished by using a single cloud provider to serve any and all applications or services that the organization decides to migrate to the cloud. Single cloud environments can utilize either private or public clouds, using whichever one better serves their current and future needs.
- Organizations with a single cloud model are employing the cloud for a single service or application, such as email, Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), or similar.
- A single cloud environment may be more fitting for smaller or less technically adept organizations that would like to gain the many benefits of the cloud without it becoming overwhelming.

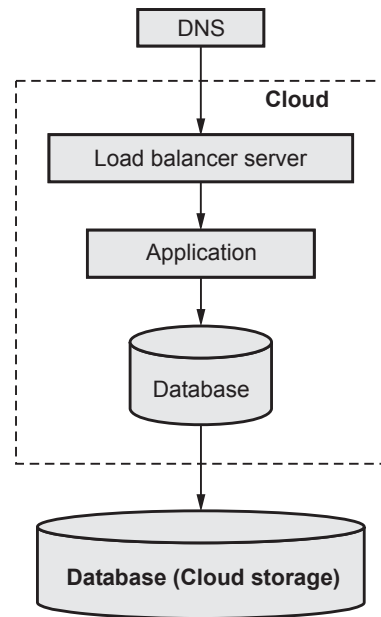


Fig. 1.9.1 Single cloud site architecture

1.9.2 Redundant 3-Tier Architecture

- Fig. 1.9.2 shows 3-tier architecture.
- Redundant three-tier architectures add another set of the same components for redundancy. Additional design components do increase complexity, but are required if designing for failover and recovery protection.
- Designing redundant infrastructures requires a well thought out plan for the components within each layer (horizontal scaling), as well as a plan for how the traffic will flow from one layer to another (vertical scaling).
- Horizontal scaling means that user can scale by adding more machines into pool of resources. For example, as traffic increases, add more web servers to take on the traffic.
- Vertical scaling means that, scale by adding more power (CPU, Memory) to an existing machine.

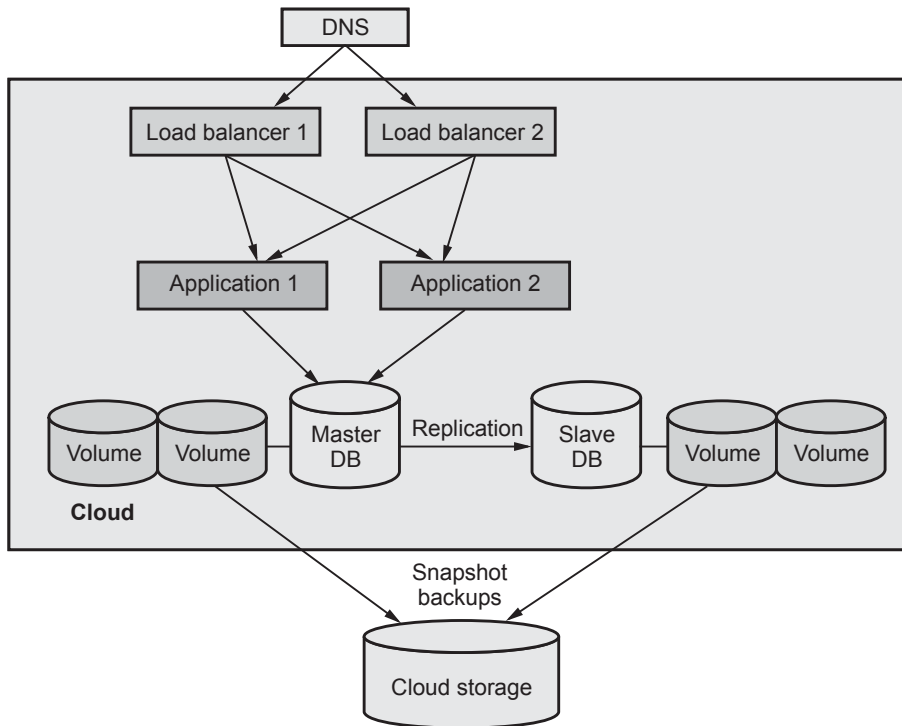


Fig. 1.9.2 3-tier architecture

- A failsafe system comprised of redundant cloud storage devices is established so that when the primary device fails, the redundant secondary device takes its place.

1.9.3 Multi-Data Centre Architecture

- Fig. 1.9.3 shows multi-data centre architecture.
- If cloud infrastructure supports multiple data centers, it's recommended that organization spread system architecture across multiple data centers to add another layer of redundancy and protection.
- Each data center in a cloud is designed to be an isolated segment inside the same geographical cloud. So if a power failure occurs in one data center, the other data centers will be unaffected.

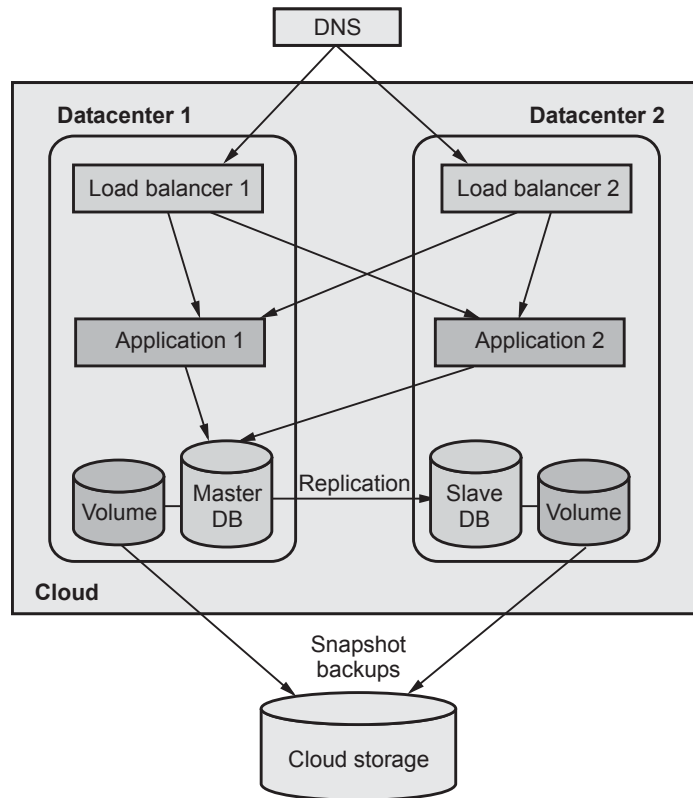


Fig. 1.9.3 Multi-data centre architecture

1.10 Cloud Deployment Models

SPPU : March-19

- Cloud deployment models refer to the location and management of the cloud's infrastructure.
- Deployment models are defined by the ownership and control of architectural design and the degree of available customization. Cloud deployment models are private public and community clouds.
- Fig. 1.10.1 shows cloud deployment model.

1. Public cloud :

- The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.
- Public cloud is a huge data centre that offers the same services to all its users. The services are accessible for everyone and much used for the consumer segment.
- Examples of public services are Facebook, Google and LinkedIn.

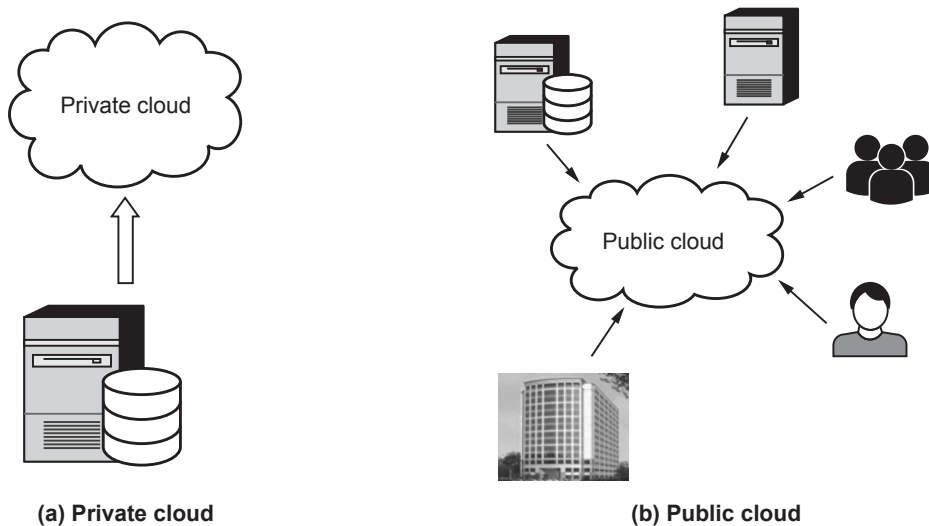


Fig. 1.10.1

- **Public cloud benefits :**

- a) Low investment hurdle : Pay for what user use.
- b) Good test/development environment for applications that scale to many servers.

- **Public cloud risks :**

- a) Security concerns : Multi-tenancy and transfers over the Internet.
- b) IT organization may react negatively to loss of control over data center function.

2. Private cloud :

- The cloud infrastructure is operated solely for a single organization. It may be managed by the organization or a third party and may exist on-premises or off-premises.

- **Private cloud benefits :**

- a) Fewer security concerns as existing data center security stays in place.
- b) IT organization retains control over data center.

- **Private cloud risks :**

- a) High investment hurdle in private cloud implementation, along with purchases of new hardware and software.
- b) New operational processes are required; old processes not all suitable for private cloud.

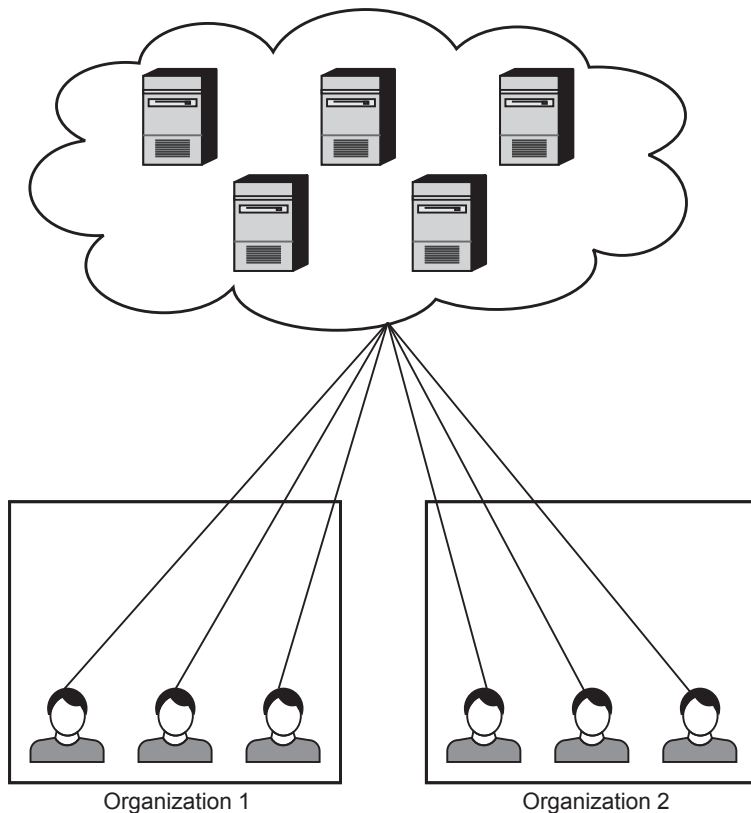


Fig. 1.10.2 Community cloud

3. Community cloud :

- The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g. mission, security requirements, policy, or compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.

4. Hybrid cloud :

- The cloud infrastructure is a composition of two or more clouds (private, community or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).
- **Hybrid cloud benefits :**
 - a) Operational flexibility : Run mission critical on private cloud, dev/test on public cloud
 - b) Scalability : Run peak and bursty workloads on the public cloud
- **Hybrid cloud risks :**
 - a) Hybrid clouds are still being developed; not many in real use

- b) Control of security between private and public clouds, some of same concerns as in public cloud

1.10.1 Difference between Public and Private Cloud

Public cloud	Private cloud
Public cloud infrastructure is offered via web applications and also as web services over Internet to the public.	Private cloud infrastructure is dedicated to a single organization.
Support multiple customer	Support dedicated customer
Full utilized of infrastructure.	Does not utilize shared infrastructure
Security is low as compared to private cloud	High level of security
Low cost	High cost
Azure, Amazon Web Services, Google App Engine and Force.com are a few examples of public clouds	An example of the Private Cloud is NIRIX's one Server with dedicated servers

Review Question

1. Explain cloud deployment models as per the NIST guidelines.

SPPU : March-19, In Sem, Marks 6

1.11 Multiple Choice Questions

Q.1 Point out the wrong statement :

- ☐ a Abstraction enables the key benefit of cloud computing : shared, ubiquitous access.
- ☐ b Virtualization assigns a logical name for a physical resource and then provides a pointer to that physical resource when a request is made.
- ☐ c All cloud computing applications combine their resources into pools that can be assigned on demand to users.
- ☐ d All of the mentioned.

Q.2 Point out the wrong statement :

- ☐ a The massive scale of cloud computing systems was enabled by the popularization of the Internet.
- ☐ b Soft computing represents a real paradigm shift in the way in which systems are deployed.

- ☐ c Cloud computing makes the long-held dream of utility computing possible with a pay-as-you-go, infinitely scalable, universally available system.
- ☐ d All of the mentioned.

Q.3 Which of the following is essential concept related to Cloud ?

- ☐ a Reliability ☐ b Productivity
- ☐ c Abstraction ☐ d All of the mentioned.

Q.4 Point out the wrong statement :

- ☐ a All applications benefit from deployment in the cloud.
- ☐ b With cloud computing, you can start very small and become big very fast.
- ☐ c Cloud computing is revolutionary, even if the technology it is built on is evolutionary.
- ☐ d None of the mentioned.

Q.5 Which of the following cloud concept is related to pooling and sharing of resources ?

- ☐ a Polymorphism ☐ b Abstraction
- ☐ c Virtualization ☐ d None of the mentioned

Q.6 CDC Stands for ____.

- ☐ a Cloud Data Computing ☐ b Cloud Data Cluster
- ☐ c Cloud Data Center ☐ d Computing Data on Cloud

Q.7 In which year IoT was introduced ?

- ☐ a 1999 ☐ b 1998
- ☐ c 1996 ☐ d 1997

Q.8 _____ is mainly used to utilize idle resources in the nodes.

- ☐ a Dedicated cluster ☐ b Enterprise cluster
- ☐ c Distributed cluster ☐ d Centralized cluster

Q.9 _____ is the entry point into the cloud for user and administrators.

- ☐ a Cloud Manager ☐ b Group Manager
- ☐ c Instance Manager ☐ d VM Manager

Q.10 The services that provides utility may directly correlate with their _____ ?

- | | |
|---|---|
| ☐ a expectations | ☐ b profit |
| ☐ c time | ☐ d satisfaction |

Q.11 The promise of _____ has raised the IT expectations of small and medium enterprises beyond measures.

- | | |
|--|--|
| ☐ a cloud computing | ☐ b cloud computing |
| ☐ c cluster computing | ☐ d client-server computing |

Q.12 Cloud supports applications _____ and resources elasticity.

- | | |
|--|--|
| ☐ a performance | ☐ b reliability |
| ☐ c scalability | ☐ d none of the above |

Q.13 _____ cloud is shared by several organizations and supports a specific community that has shared concerns.

- | | |
|------------------------------------|--------------------------------------|
| ☐ a Private | ☐ b Public |
| ☐ c Managed | ☐ d Community |

Q.14 _____ of grids/clouds to offer standard interfaces for dynamically scalable services delivery in their products.

- | | |
|--|--------------------------------------|
| ☐ a Vendors | ☐ b Consumers |
| ☐ c Integrators | ☐ d Providers |

Q.15 VAN Stands for _____ .

- | | |
|--|---|
| ☐ a Virtual Application Network | ☐ b Virtual Acceptable Network |
| ☐ c Virtual Admission Network | ☐ d Virtual Area Network |

Q.16 In what type of cluster, the nodes are closely packaged in one or more racks sitting in a room, and the nodes are not attached to peripherals ?

- | | |
|--|--|
| ☐ a Compact | ☐ b Slack |
| ☐ c loosely coupled | ☐ d Tightly coupled |

Q.17 Virtualization is a computer architecture technology by which multiple _____ are multiplexed in the same hardware machine.

- | | |
|--|--|
| ☐ a virtual memory | ☐ b virtual machines |
| ☐ c physical machines | ☐ d virtual machine monitor |

Q.18 Which of the following is not VI manager ?

- ☐ a Apache VCL ☐ b App Logic
☐ c Google VI ☐ d Nimbuz 3

Q.19 According to Infosys how many steps are there in the migration model ?

- ☐ a 4 ☐ b 5
☐ c 6 ☐ d 7

Q.20 _____ is the process of transferring data between storage types, formats or systems.

- ☐ a Data mediation ☐ b Data integrity
☐ c Data modification ☐ d Data migration

Q.21 Which of the following is not principle of the cloud ?

- ☐ a Federation ☐ b Non independent
☐ c Trust ☐ d Isolation

Q.22 _____ is a directory on the cluster node where a VM is running.

- ☐ a Virtualization ☐ b KVM
☐ c Virtual Machine Directory ☐ d VMware

Answer keys for Multiple Choice Questions :

Q.1	c	Q.2	b	Q.3	c	Q.4	a
Q.5	c	Q.6	c	Q.7	a	Q.8	b
Q.9	a	Q.10	a	Q.11	b	Q.12	c
Q.13	d	Q.14	a	Q.15	d	Q.16	a
Q.17	b	Q.18	b	Q.19	d	Q.20	d
Q.21	a	Q.22	b				



Notes

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